

Block-I

UNIT-3

TRANSFORMATON OF EQUATIONS

Objectives

After completion of this unit, students will be able to

- ★ understand the concept of transformation of equations.
- ★ identify the reciprocal equations and solve the equations.

3.1 Transformation of Equations:

It is possible to transform the given equation into another whose roots bear with the roots of the original equation.

(i) **Roots with signs changed:**

To transform an equation into another whose roots are numerically the same as those of the given equation but opposite in sign. To get this result, we have to substitute $-x$ for x in the given equation.

(ii) **Roots multiplied by a given number:**

To transform an equation into another whose roots are m times that of the given equation. To effect this transformation, we have to multiply the successive terms beginning with the second by $m, m^2, m^3, \dots, m^n$

This transform is useful for the purpose of removing the coefficient of the first term of an equation when it is other than unity and generally for removing the fractional coefficients from the equation.

Example 3.1. Find the equation whose roots are the roots of $x^5 + 6x^4 + 6x^3 - 7x^2 + 2x - 1 = 0$ with the signs changed

Solution:

$$\text{Given } f(x) = x^5 + 6x^4 + 6x^3 - 7x^2 + 2x - 1 = 0$$

By substituting $x = -x$, we get,

$$(-x)^5 + 6(-x)^4 + 6(-x)^3 - 7(-x)^2 + 2(-x) - 1 = 0$$

$$\text{i.e., } -x^5 + 6x^4 - 6x^3 - 7x^2 - 2x - 1 = 0$$

By multiplying with -1 on both sides we get

$$x^5 - 6x^4 + 6x^3 + 7x^2 + 2x + 1 = 0$$

Example 3.2. Change the sign of the roots of the equation

$$x^7 + 4x^5 + x^3 - 2x^2 + 7x + 3 = 0$$

Solution:

$$\text{Given } x^7 + 4x^5 + x^3 - 2x^2 + 7x + 3 = 0$$

By substituting $x = -x$, we get,

$$(-x)^7 + 4(-x)^5 + (-x)^3 - 2(-x)^2 + 7(-x) + 3 = 0$$

$$\text{i.e., } -x^7 - 4x^5 - x^3 - 2x^2 - 7x + 3 = 0$$

By multiplying with -1 on both sides we get

$$x^7 + 4x^5 + x^3 + 2x^2 + 7x - 3 = 0$$

Example 3.3. Change the equation $2x^4 - 3x^3 + 3x^2 - x + 2 = 0$ into another the coefficient of whose highest term will be unity.

Solution: Given equation is $2x^4 - 3x^3 + 3x^2 - x + 2 = 0$

Since the coefficient of the highest term is 2, we can multiply the roots by

2. Thus, the transformed equation becomes

$$2x^4 - (2^1)3x^3 + (2^2)3x^2 - (2^3)x + (2^4)2 = 0$$

$$\text{i.e., } 2x^4 - 6x^3 + 12x^2 - 8x + 32 = 0$$

$$\text{i.e., } x^4 - 3x^3 + 6x^2 - 4x + 16 = 0$$

Example 3.4. Remove the fractional coefficients from the equation

$$x^3 - \frac{1}{4}x^2 + \frac{1}{3}x - 1 = 0.$$

Solution: To transform the equation into another whose roots are multiplied by m , therefore we get $x^3 - \frac{m}{4}x^2 + \frac{m^2}{3}x - m^3 = 0$

By substituting values for m , the result $\frac{m}{4}$ and $\frac{m^2}{3}$ will be an integer.

So, when $m = 12$, $\frac{m}{4}$ and $\frac{m^2}{3}$ will be integer.

∴ Multiply the roots by 12, we get,

$$\begin{aligned} x^3 - \frac{12}{4}x^2 + \frac{12^2}{3}x - 12^3 &= 0 \\ \Rightarrow x^3 - 3x^2 + 48x - 1728 &= 0 \end{aligned}$$

Example 3.5. Remove the fractional coefficients from the equation

$$2x^3 + \frac{3}{2}x^2 - \frac{1}{8}x - \frac{3}{16} = 0.$$

Solution: Given $2x^3 + \frac{3}{2}x^2 - \frac{1}{8}x - \frac{3}{16} = 0$ (3.1)

In order to remove the fractional coefficients first change the coefficient of the highest degree to unity by dividing by 2, we get

$$x^3 + \frac{3}{4}x^2 - \frac{1}{16}x - \frac{3}{32} = 0$$

Multiply the roots by m , we get $x^3 + \frac{3m}{4}x^2 - \frac{m^2}{16}x - \frac{3m^3}{32} = 0$

When $m = 4$, then $\frac{3m}{4}$, $\frac{m^2}{16}$, $\frac{3m^3}{32}$ will be integer and hence multiply the roots by 4

$$\begin{aligned} \therefore \text{Equation (3.1)} &\Rightarrow x^3 + \frac{(3)(4)}{4}x^2 - \frac{(4)^2}{16}x - \frac{(3)(4^3)}{32} = 0 \\ &\Rightarrow x^3 + 3x^2 - x - 6 = 0 \end{aligned}$$

Check Your Progress:

1. Change the sign of the roots of the equation

$$x^7 + 4x^5 + x^3 - 2x^2 + 7x + 3 = 0$$

2. Remove the fractional coefficients from the equation

$$x^3 + \frac{1}{4}x^2 - \frac{1}{16}x + \frac{1}{72} = 0$$

3. Transform the equation $3x^3 + 4x^2 + 5x - 6 = 0$ into one in which the coefficient of x^3 is unity and all the coefficients are integral.

4. Remove the fractional coefficients from the equation

$$x^3 + \frac{3}{2}x^2 + \frac{5}{18}x + \frac{1}{108} = 0$$

Answer:

1. $x^7 + 4x^5 + x^3 + 2x^2 + 7x - 3 = 0$

2. $x^3 + 3x^2 - 9x + 24 = 0$

3. $x^3 + 4x^2 + 15x - 54 = 0$

4. $x^3 + 9x^2 + 10x + 2 = 0$

3.2 Reciprocal Equation:

A reciprocal equation is one which is unaltered when x is replaced by $\frac{1}{x}$.

Note 3.1.

1. If α is a root, then $\frac{1}{\alpha}$ is also a root of the equation.

2. In the reciprocal equation, $a_0 = \pm a_n$, $a_1 = \pm a_{n-1}, \dots$

Hints for solving the reciprocal equations.

Type	Degree of $f(x) = 0$	Signs of a_0 & a_n	Result
Type I (Standard Type)	Even	Same	If the degree of $f(x) = 0$ is 4, then divide the given equation by x^2 (If it is 6, then divide by x^3)
Type II	Even	Opposite	$x = 1$ and $x = -1$ are the roots
Type III	Odd	Same	$x = -1$ is one of the root
Type IV	Odd	Opposite	$x = 1$ is one of the root

Note 3.2. In Type 1, for the fourth degree equation, **after dividing by x^2** , we can put

$$(i) \quad x + \frac{1}{x} = z \quad \text{and} \quad (ii) \quad x^2 + \frac{1}{x^2} = z^2 - 2$$

Example 3.6. Solve $4x^4 - 20x^3 + 33x^2 - 20x + 4 = 0$ (Standard Type)

Solution:

$$\text{Given equation is } 4x^4 - 20x^3 + 33x^2 - 20x + 4 = 0 \quad (3.2)$$